Retail Sales & Business Performance Analysis

End-to-end Data Analyst Portfolio Project

Tools	Excel | SQL | Python (Pandas) | Power BI
Project Type	Synthetic retail business analysis for portfolio/interview practice
Target Role	Data Analyst Intern / Fresher Data Analyst
Primary Goal	Convert raw sales data into KPIs, trends, analysis and business insights

GitHub note: This PDF is the project documentation. Upload it together with the included CSV, SQL and Python files so recruiters can inspect the work.

1. Project Overview

This project demonstrates a practical data analyst workflow: collect/inspect data, clean it, validate quality, analyze business performance, calculate KPIs, create dashboards, and summarize findings in simple business language. The dataset in this package is synthetic and is intended for portfolio and interview practice rather than representing Crossing Infotech internal data.

Business Questions

- What are total sales, orders, customers and average order value?
- Which regions and product categories generate the most sales?
- Which products are the top performers?
- How do sales change month by month?
- Where are there potential dips or opportunities to investigate?

2. Dataset Structure

Order_ID	Order_Date	Customer_ID	Region	Product	Category	Quantity	Unit_Price	Discount	Sales
O10001	2026-02-27	C1032	East	Laptop	Electronics	2	603.14	0	1206.28
O10002	2026-02-14	C1005	West	Water Bottle	Accessories	1	16.54	0.02	16.21
O10003	2026-09-16	C1072	North	Water Bottle	Accessories	2	18.78	0.1	33.8
O10004	2026-08-03	C1076	West	Office Chair	Furniture	3	191.14	0	573.42
O10005	2026-03-23	C1044	West	Pen Set	Stationery	3	9.31	0.05	26.53
O10006	2026-02-22	C1013	West	Smartphone	Electronics	3	449.19	0.1	1212.81

Data dictionary: Order_ID = unique order; Order_Date = transaction date; Customer_ID = customer; Region = sales region; Product = item; Category = product group; Quantity = units; Unit_Price = unit price; Discount = discount rate; Sales = final sales value.

3. Excel Work

In Excel, load the CSV and perform the following steps:

- Check blank cells and duplicate Order_ID values.
- Create a cleaned data table and standardize date/number formats.
- Use SUMIFS/COUNTIFS or Pivot Tables for sales by region, category and month.
- Use XLOOKUP/VLOOKUP when a separate product master or target table is introduced.
- Create Pivot Charts for monthly sales, regional performance and top products.
- Build a one-page management report with KPI cards and 2-4 supporting charts.

Example formulas

```
Total Sales      = SUM(Sales_Column)
Order Count      = COUNTA(Order_ID_Column)
Unique Customers = COUNTA(UNIQUE(Customer_ID_Column))
Region Sales     = SUMIFS(Sales_Column, Region_Column, "West")
```

4. SQL Analysis

The included SQL file contains reusable business questions. Examples:

```
SELECT Region, ROUND(SUM(Sales),2) AS Sales
FROM retail_sales
GROUP BY Region
ORDER BY Sales DESC;

SELECT Product, ROUND(SUM(Sales),2) AS Sales
FROM retail_sales
GROUP BY Product
ORDER BY Sales DESC
LIMIT 10;

SELECT DATE_FORMAT(Order_Date, '%Y-%m') AS Month,
      ROUND(SUM(Sales),2) AS Sales
FROM retail_sales
GROUP BY Month
ORDER BY Month;
```

Interview talking point: SQL is used here to answer exact business questions quickly and reproducibly, especially when a report requires filtering, grouping, aggregation or joining multiple tables.

5. Python / Pandas Work

Python is used to automate repeatable checks and analysis. The included script loads the CSV, checks missing values and duplicates, calculates KPIs, creates monthly/region/product summaries and prepares a trend chart.

```
df = pd.read_csv("retail_sales.csv")
df["Order_Date"] = pd.to_datetime(df["Order_Date"])

print(df.isnull().sum())
print("Duplicates:", df.duplicated().sum())

total_sales = df["Sales"].sum()
avg_order_value = df["Sales"].mean()

df["Month"] = df["Order_Date"].dt.to_period("M").astype(str)
monthly = df.groupby("Month", as_index=False)["Sales"].sum()
```

Why use Python? It makes repetitive cleaning and calculations easier to rerun on a new CSV rather than repeating every step manually.

6. Power BI Dashboard Plan

Build a single-page interactive dashboard with these sections:

- Top KPI cards: Total Sales, Total Orders, Unique Customers, Average Order Value.
- Line chart: Monthly Sales Trend.
- Bar chart: Sales by Region.
- Bar chart: Sales by Category.
- Top 10 Products table or bar chart.
- Slicers: Date, Region and Category.

Total Sales 236,101.42	Total Orders 500	Customers 80	Average Order Value 472.20

7. Example Findings from the Synthetic Dataset

These figures come only from the sample dataset included in this project package. They are not claims about any real company.

Metric	Result	Sales
Highest-sales region	West	70,937.04
Highest-sales category	Electronics	177,082.93
Highest-sales product	Laptop	73,152.56
Highest-sales month	2026-02	34,416.91

How to present an insight: state the metric, describe the trend, then explain what action or follow-up question it suggests. Avoid claiming a cause unless the data supports it.

8. GitHub Upload Structure

```
Retail-Sales-Business-Analysis/
|-- retail_sales.csv
|-- retail_sales_analysis.sql
|-- data_cleaning_analysis.py
|-- Retail_Sales_Business_Analysis_Project.pdf
|-- README.md
`-- data_dictionary.txt
```

Recommended GitHub repository name: retail-sales-business-analysis

Recommended project headline: Retail Sales & Business Performance Analysis | Excel, SQL, Power BI, Python

Upload order: create a new GitHub repository -> add the project files -> commit changes -> open the README -> verify that the files appear correctly.

9. Interview Explanation

30-second answer: “I built an end-to-end retail sales analysis project using Excel, SQL, Python and Power BI. I started by checking data quality and cleaning the dataset, then used SQL and Pandas to calculate KPIs and identify trends. Finally, I designed a Power BI dashboard for sales, orders, customers, regions and products, and summarized the findings in business-friendly language.”

Questions you should be ready for

- Why did you choose these KPIs?
- How did you check for duplicates and missing values?
- Why use SQL when you already have Excel?
- What would you do if Power BI and SQL showed different totals?
- Which business question does your dashboard answer?